

A-level

ENVIRONMENTAL SCIENCE

Paper 2 - Generated practice paper Set 28

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	• Success: Cactoblastis moth introduced from South America controlled prickly pear cactus in Australia; Encarsia parasitic wasps control greenhouse whitefly • Problems: cane toads introduced to Australia to control sugar cane beetles did not control them but ate many other species and are toxic to predators; rosy wolfsnail introduced to Polynesia to control African land snails ate endemic Partulid snails (48 species extinct); harlequin ladybirds introduced to Europe eat native ladybirds • Lesson: control agents should be specialist feeders on the pest	4
01.2	• Release an artificial sex scent attracting males (or females) of the pest species • Used to monitor when the pest arrives so pesticide is timed correctly, or (in enclosed greenhouses) to trap enough of one sex to prevent breeding	2
01.3	• Regular low doses kill only the most sensitive bacteria so resistant strains survive and multiply • Agricultural use exceeds medical use, so resistance spreads • Resistant zoonotic bacteria (E. coli, Salmonella, Campylobacter) can transfer to humans and are harder to treat; hence reduced use of antibiotics is a sustainability goal	3
01.4	• Grow nectar-rich flowering plants alongside crops; restrict pesticides harmful to bees / spray when not in flower; introduce bee hives	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 316-320; AQA spec 3.5.1.1.	

Question	Marking guidance	Marks
02.1	• Selective breeding: parents chosen for desirable genetic characteristics (Limousin muscle growth, Merino wool, Cheviot hardiness); combines genes but characteristics of offspring not fully predictable; many generations needed; risk of inbreeding • Asexual reproduction (cuttings, runners, tissue culture, cloning): offspring genetically identical to the parent so characteristics are predictable and survival high; but no variation so no improvement and vulnerability to the same disease; cloning of animals (nucleus transfer into an enucleated egg, surrogate) still developing • Cross-breeding: two breeds combined to give a mix of characteristics with hybrid vigour (heterosis) and lower inbreeding risk, eg Zebu (heat tolerant) x Ayrshire (high milk yield)	6
02.2	• AI: semen from a male with desirable characteristics can fertilise many more females than natural mating; frozen semen stored and transported easily and used after the male dies • Embryo transfer: FSH stimulates a superior female to release many eggs, which are fertilised (IVF) with sperm from a superior male • Each embryo is implanted in a different surrogate female • So the donor produces many offspring per year rather than one per pregnancy	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 321-323; AQA spec 3.5.1.2.	
03.1	• Island communities evolved from few colonising species in isolation, often without mammal predators or large herbivores • So indigenous species lack adaptations (eg ground-nesting birds, flightlessness) to survive introduced cats, rats, pigs, goats	2
03.2	• Competitor: grey squirrel (North America) out-competes red squirrel - digests acorns, larger, wins nest sites; rhododendron shades ground and releases toxins preventing native regrowth • Predator: American mink (escaped fur farms) reduced water vole range; cane toads in Australia; cats / rats / pigs on islands eating ground-nesting birds; Nile perch in Lake Victoria ate cichlids to extinction • Pathogen: squirrel pox virus carried by grey squirrels kills red squirrels; crayfish plague carried by signal crayfish kills white-clawed crayfish; Dutch elm disease, ash dieback, sudden oak death imported on plants / soil	6
03.3	• Closely related introduced species (sika deer; domestic cats) cross-breed producing fertile hybrids • The natural gene pool is changed / diluted with genes it would not naturally contain, so the pure indigenous species may disappear	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 24-25; AQA spec 3.1.2.2.	
04.1	• Harvest rate below the maximum sustainable yield; only economically valuable species (mahogany) selectively logged, not clear-felled • Trees felled only where other mahoganies stand upwind so seeds recolonise the clearing; large seed trees and small trees left to grow • Buffer zones with no harvesting or machinery near rivers / wetlands • Restricted machinery and planned track routes re-used to avoid damaging roots of retained / rare / fruit trees • Each felled tree tagged and tracked to the retailer for FSC accreditation; rangers control illegal logging; tree nurseries replant over-exploited and ecologically important species	5

Question	Marking guidance	Marks
04.2	• The reserve removed traditional uses (fuelwood, timber, farming clearings, hunting, medicines) so people would otherwise resent / ignore it • Employment as rangers, forestry and tourism workers; ecotourists use local restaurants, produce and souvenirs; offcuts given to local carvers • Benefits give the community an incentive to protect the forest; carbon sequestration payments from companies fund management	3
04.3	• Removing individual trees leaves small clearings so remaining populations survive nearby and the forest structure / microclimate is kept; timber resource used over a longer period • Disadvantage: more labour-intensive and expensive; more tracks needed per tree	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 369, 386-387; AQA spec 3.5.3.3.	
05.1	• k-selected: very slow growth, does not breed until 30, few eggs, so the population cannot replace losses • Fish reach catchable size (20) before they breed (30) so fishing can remove the entire future breeding population • Deep-water aggregations easily located and caught by trawling • Recovery takes decades so fisheries have had to be closed	4
05.2	• Herring breed at 2 years - the same age they become catchable - so there is always a breeding population and many young • Cod are catchable at 2 but do not mature until 4, so heavy fishing removes most fish before they breed	2
05.3	• Larger numbers of smaller / younger fish are caught, or bigger boats, nets and sonar increase catch effort • Masks the decline of the breeding population until a sudden collapse	2
05.4	• Neither is a target species but both are caught in demersal trawls; skate now extinct in the Baltic and critically endangered elsewhere • Very late maturity (skate 11 years, Greenland shark 100-150 years, living over 400) and few young mean even low by-catch mortality exceeds replacement	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 341-342, 348-349; AQA spec 3.5.2.2.	
06.1	• General physiological solvent: most biological reactions take place between substances dissolved in water • Transport medium in blood / sap carrying gases, sugars, amino acids, nutrients, wastes • High specific heat capacity moderates temperature changes; evaporation absorbs heat (cooling) • Anomalous expansion: ice floats so water bodies do not freeze solid; absorbs UV protecting early marine life; provides aquatic habitats	4

Question	Marking guidance	Marks
06.2	- From about 2.7 billion years ago photosynthetic archaea / bacteria (then algae and plants) released oxygen; at first it reacted with dissolved iron (forming iron ore deposits) then built up in the oceans and atmosphere (from about 2.45 billion years ago) - Oxygen in the stratosphere absorbed UV, forming ozone (O₃) in a dynamic equilibrium of O, O₂ and O₃ - The ozone layer reduced UV reaching the surface so organisms could survive out of the water - Oxygen allowed aerobic respiration releasing more energy so complex multicellular organisms evolved; many anaerobic archaea died out	4
06.3	- Photosynthesis removed CO₂ (a greenhouse gas) from the atmosphere, storing carbon in sediments, carbonate rocks and fossil fuels - This reduced infrared retention, offsetting the increasing brightness of the Sun (about 10% per billion years) so temperatures did not rise excessively	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 7-8; AQA spec 3.1.1.1-3.1.1.2.	
07.1	- Upwelling currents bring nutrients from the seabed to surface waters 24-hour daylight in summer allows a huge phytoplankton bloom - Algae also grow under sea ice where krill shelter and feed; krill support most of the food web	3
07.2	- Krill is the main food of baleen whales, penguins, seals and fish so over-harvesting removes food from most of the web - Toothfish overfishing removes the food of sperm whales and elephant seals - Long-lining for toothfish catches and drowns albatrosses on the hooks (by-catch)	3
07.3	- Ice on land stores 70% of the Earth's fresh water, keeping sea levels low - High albedo of ice reflects sunlight, limiting warming (reduction in ice area causes further warming) - Carbon sequestration: carbon in algae eaten by krill sinks to the seabed in krill faeces	3
07.4	Stronger winds spreading ice; more snowfall from warmer, moister air freezing onto slush; fresh meltwater from land forming a surface layer that freezes more easily (long-term trend still expected to be a decline)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 67-71; AQA spec 3.1.2.3.4.	
08.1	- Irrigation depletes rivers and aquifers (eg Aral Sea, Ogallala); increases evaporation - Soil erosion and compaction reduce water retention and infiltration so runoff and river flow fluctuations increase - Evapotranspiration changes: increased in irrigated arid areas, decreased where forests were cleared	3
08.2	- Seasonal / local: avoids energy for heating, lighting greenhouses out of season and long-distance transport (food miles) and the pollution this causes - Organic: uses natural processes (rotation, legumes, manure, biological control, predator habitats) instead of artificial fertilisers and pesticides, so less pollution, energy use and harm to non-target species; free-range and Fairtrade also accepted for welfare / social reasons	4

Question	Marking guidance	Marks
08.3	• Food shortages after 1945 (dependence on US food aid); farmers could not afford loans to invest so grants funded machinery, drainage, hedgerow removal, liming, improved livestock • Guaranteed market / price support: government bought surpluses to hold prices up (and sold stores in poor years) giving farmers security to increase output • Consequences: loss of hedgerows and wetlands, intensification, pesticide and fertiliser pollution; later food surpluses (butter mountains) that could not be sold without harming LEDC producers	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 329-333; AQA spec 3.5.1.3-3.5.1.4.	
09.1	• $\Sigma n(n - 1) = (40 \times 39) + (30 \times 29) + (20 \times 19) + (10 \times 9) = 1560 + 870 + 380 + 90 = 2900$ • $N(N - 1) = 100 \times 99 = 9900$ • $D = 9900 / 2900 = 3.41$ (accept 3.4)	3
09.2	• Woodland A has a higher biodiversity (greater variety and more even abundance) than woodland B • The index does not identify which species are present / a rare or important species may be missed; depends on identification skill and sampling method	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 402; AQA spec 3.7.2.3.	

Question	Marking guidance	Marks
10.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Resources: wood, fibres (cotton, wool, silk), oils, fuels (wood, biofuels), food; new food species (morama bean, perennial cereals, eland, bison) better adapted to local conditions • Biomimetics: vehicle design, architecture, structures, adhesion, materials, ultrasound • Medicines from plants and animals (examples); most species unstudied - argument for habitat conservation • Physiological research using wildlife species; wildlife as pest control agents (ladybirds, Encarsia, Cactoblastis, Bt) • Genetic resources: crop wild relatives, Vavilov centres, gene pools • Ecosystem services: atmospheric composition, hydrological and biogeochemical cycles, soil maintenance, pollination, seed dispersal, habitat provision • Ethical / aesthetic / recreational arguments; the unknown value of species not yet investigated (precautionary principle) • Evaluation: which arguments are most persuasive to decision makers; economic valuation vs intrinsic value; judgement	9
10.2	• The variety and abundance of life (species, genes, habitats) in an area / on Earth	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 10-19; AQA spec 3.1.2.1.	

Question	Marking guidance	Marks
11.1	Discuss how ecological monitoring and the development of new technologies have improved the planning and evaluation of conservation strategies. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Why monitor: species present, population size, density, distribution, survival, age structure, breeding rates, movements; baseline and evaluation; Red List categorisation and quotas rely on it • Standard techniques and their limitations: quadrats, transects, nets, traps, Lincoln index, Simpson's index, DAFOR; sampling design (random / systematic, size, number, timing, standardisation) • New technologies: satellite / radio / GPS / acoustic tracking, geolocators, camera traps, CCTV, image recognition databases, DNA databases, eDNA, sonograms, drones / ROVs / AUVs, satellite sensors, data loggers • How they improve validity: representative data over large areas and times, less disturbance, individual identification, detection of hidden species, objectivity • Examples: penguin colonies from satellites, tiger camera traps, ivory DNA, MCZ monitoring, released animals tracked • Limitations: cost, battery / tag weight, data handling, ethics, still need ground truthing • Judgement Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 393-394, 400-407; AQA spec 3.1.2.3.5-3.1.2.3.6, 3.7.	25

Question	Marking guidance	Marks
11.2	Natural systems have remained sustainable over very long periods. Discuss how the features of natural systems could be copied to make human systems more sustainable. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Dynamic equilibria and negative feedback in climate, hydrological cycle and populations; positive feedback and tipping points caused by humans • Energy: solar-driven, low energy density, low temperature (photosynthesis, enzymes, nitrogen fixation vs Haber process); renewable energy, heat pumps, bioleaching, fuel cells • Materials: cycles, few abundant elements, biodegradable non-toxic wastes vs fossil fuels and dispersed minerals; recycling, Cradle to Cradle, industrial symbiosis • Diversity and stability: rainforests / reefs vs agroecosystems; diverse energy mixes • Optimum vs maximum production; supporting ecosystems in agriculture, forestry, fisheries, urban design • Case studies and metrics; limits and judgement Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 370-379; AQA spec 3.6.1-3.6.4.	25